

Fishing for Sustainability

Activity Procedure

Wed Mar 25, 2026

Introduction

In this activity, you will simulate a **shared fishery**—a resource that everyone can access but no one owns.

You should work in **groups of ~4-6 people** for the entire exercise.

You will participate in three short rounds, each with different rules. Your goal is to observe how:

- Individual decisions affect the **entire population**
- Different management strategies influence **sustainability**
- Human behavior creates challenges in managing **common-pool resources**

Your Role

You are an independent fisher.

- Your goal is to **maximize your personal catch**
- However, the fish population is **shared by everyone**
- Your decisions will affect both **your outcome** and the **group outcome**

The System (Applies to All Rounds)

- The lake begins with **30 fish**
- After each round:
 - The remaining fish **reproduce (double)**
 - The population cannot exceed **30 fish** (carrying capacity, K)

What Changes Each Round

Round	System	Key Idea
1	Open Access	No rules or communication
2	Fixed Quota	Group agrees on a total limit
3	Fixed Effort	Each person follows the same individual limit

Key Concepts

You should be able to apply these ideas after the activity:

Tragedy of the Commons When individuals acting in their own interest deplete a shared resource

Social Dilemma When what is best for individuals conflicts with what is best for the group

Carrying Capacity (K) The maximum population size the environment can sustain (here, 30 fish)

Maximum Sustainable Yield (MSY) The largest long-term harvest that does not reduce the population over time

Round 1: Open Access

Test whether a **per-person rule** improves sustainability, fairness, and compliance.

You are an independent fisher trying to **maximize your personal catch**. Unlike Round 2, this system sets a **strict individual limit** that everyone must follow.

- 1 bowl = **lake** (starts with **30 “fish”**)
- 8 cups = **boats** (one per student)

! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).

If the fishery **collapses**, no fish remain and **no one gains anything further from this round**.

Communication Rules

Read these aloud & follow exactly

- **Do NOT talk**, signal, make eye contact, or coordinate in any way.
- **You are independent fishers**. No communication is permitted **until after** this hand-out is collected.

! You may communicate:

Nothing at all. No talking, signaling, or writing to each other.

Reminder: Some fishers may try to maximize their catch. Others may try to conserve. You do **not** know what others will do.

Procedure

- (1) **Secret Choice (0–6):** Privately decide how many fish to take (0–6). Write the number on your paper; **do not show** anyone.
- (2) **Simultaneous Harvest:** When the instructor says “**Fish!**” everyone takes their chosen fish at the same time.
- (3) **If total fish taken > fish available:** The fishery **collapses** (0 fish remain). **No recovery. Stop.**
- (4) **If fish remain:** Instructor counts the remaining fish, then **doubles** them (up to 30).
- (5) **Record** your results in the data table for this round and answer the reflection questions **before** you proceed.

Round 2: Fixed Quota

Test whether a **group quota** improves sustainability—and whether people will follow it.

You are an independent fisher trying to **maximize your personal catch**. The class may agree on a quota, but your **individual payoff still depends on how many fish you take**.

- Reset lake to **30 fish**
- Same 8 boats (cups)

! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).
If the class exceeds the quota, the fishery **collapses** (0 fish remain).

There is no recovery for this round.

Communication Rules

Read these aloud & follow exactly

- **Discussion Allowed (2 minutes ONLY):** You may talk to agree on a **total class quota** (e.g., “We will catch at most 16 fish”).
- After the 2-minute discussion ends:
 - **No further communication** (no talking, signaling, or eye contact)
 - All decisions are **private**

! You may communicate (for 2 minutes only):

- Propose and agree on a **single total class quota**

You may NOT communicate (after time ends):

- Your personal intended catch
- Any signals or coordination during harvesting

Reminder: The quota is a **group agreement**, not automatically enforced. Individuals can still choose how much to take.

Procedure

- (6) **Set the Quota (group, 2 minutes):** Agree on a total class quota: **Class quota = _____ fish**
- (7) **Secret Choice (0–6):** After discussion ends, privately choose how many fish to take (0–6). **Do not reveal your number.**
- (8) **Simultaneous Harvest:** When the instructor says “**Fish!**” everyone takes their chosen fish at the same time.

(9) **Check Outcome:**

- If **total class catch** $>$ **quota** $\rightarrow$ fishery **collapses** (0 fish remain). **Stop.**
- If **total class catch** $\leq$ **quota** $\rightarrow$ instructor counts remaining fish, then **doubles** them (up to 30).

(10) **Record** your results in the data table for this round and answer the reflection questions **before** you proceed.

Round 3: Fixed Effort

Test whether a **per-person rule** improves sustainability, fairness, and compliance.

You are an independent fisher trying to **maximize your personal catch**. Unlike Round 2, this system sets a **strict individual limit** that everyone must follow.

- Reset lake to **30 fish**
- Same 8 boats (cups)

! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).

Key Difference from Round 2:

The rule applies **equally to each individual** and is **fully enforced**.

Communication Rules

Read these aloud & follow exactly

- **Discussion Allowed (2 minutes):** You may talk to agree on a **per-person limit** (e.g., “2 fish per fisher”).
- After the 2-minute discussion ends:
 - **No further communication** (no talking, signaling, or eye contact)
 - Everyone must follow the agreed rule exactly

! You may communicate (for 2 minutes only):

- Propose and agree on a **per-person limit**

You must follow:

- Take **exactly** the agreed number of fish
- Do **not** exceed or go below the limit

You may NOT:

- Signal, negotiate, or change behavior during harvesting

Reminder: This rule is **enforceable**—everyone is held to the same limit. You cannot increase your payoff by taking more than others.

Procedure

- (11) **Set the Per-Person Limit (group, 2 minutes):** Per-person limit = _____ fish
- (12) **Harvest Exactly the Limit:** When the instructor says “**Fish!**” each fisher takes **exactly** the agreed limit.

- (13) **Reproduction:** Instructor counts remaining fish, then **doubles** them (up to 30).
- (14) **Record** your results in the data table for this round and answer the reflection questions.